

**ADVANCING ENVIRONMENTAL SUSTAINABILITY THROUGH SATELLITE IMAGERY
BASED ESTIMATION OF PM_{2.5}**

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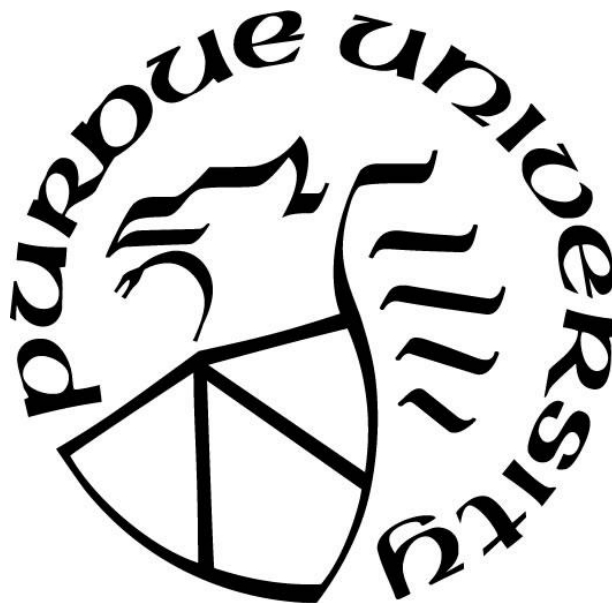
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*To my parents for their endless sacrifices, love, and belief in me, and to my friends and mentors
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ABSTRACT

Fine particulate matter ($\text{PM}_{2.5}$) is a major environmental and public health concern, contributing to respiratory and cardiovascular diseases and reduced quality of life, particularly in urban and industrial regions. Effective air quality monitoring is essential for sustainable urban planning and environmental decision-making. However, traditional monitoring approaches rely on ground-based sensor networks that are spatially sparse, unevenly distributed, and costly to maintain. These limitations make it difficult to capture fine-grained spatial and temporal variations in $\text{PM}_{2.5}$ concentrations at regional and global scales.

Satellite remote sensing offers a scalable and cost-effective alternative by providing consistent observations of land surface and atmospheric conditions over large geographic areas. I propose a simple framework for estimating $\text{PM}_{2.5}$ concentrations using high-resolution Sentinel-2 satellite imagery combined with geographic location and temporal information. To mitigate noise, missing values and short-term fluctuations in ground-based air quality measurements, $\text{PM}_{2.5}$ observations are temporally aggregated, enabling the model to learn more stable seasonal and regional trends.

We train a neural network that integrates satellite-derived image features with spatial and temporal context to model the complex relationships underlying $\text{PM}_{2.5}$ variability. The proposed framework supports simple scalable and sustainable air quality monitoring by enabling the generation of dynamic $\text{PM}_{2.5}$ maps, which can inform environmental assessment, sustainable urban planning and operational decision-making aimed at improving public health outcomes.

1. Introduction

Fine particulate matter with an aerodynamic diameter less than 2.5 micrometers ($PM_{2.5}$) is among the most harmful air pollutants due to its ability to penetrate deep into the respiratory system and enter the bloodstream. Long term exposure to elevated $PM_{2.5}$ concentrations has been strongly associated with respiratory and cardiovascular diseases, premature mortality and reduced quality of life. As urbanization and industrial activity continue to intensify worldwide, the ability to monitor and manage $PM_{2.5}$ levels has become a critical component of environmental sustainability, public health protection and sustainable urban development, supporting long-term strategies aimed at reducing population exposure and improving overall well-being.

Conventional air quality monitoring systems rely primarily on ground-based sensor networks that provide accurate measurements at fixed locations. While these stations offer reliable point level observations, they are often sparsely distributed, unevenly located and costly to deploy, operate and maintain. As a result, large geographic regions particularly in rapidly urbanizing areas and low and middle-income regions lack sufficient monitoring coverage, limiting the sustainability of long-term air quality management efforts and constraining data driven environmental planning and policy evaluation.

Satellite remote sensing has emerged as a valuable complement to ground-based monitoring by offering consistent observations over broad spatial extents. Many existing satellite-based approaches for estimating $PM_{2.5}$ concentrations rely on aerosol optical depth (AOD) products and their integration with meteorological variables and chemical transport models. Although these methods have demonstrated effectiveness at regional and global scales, they often face challenges related to cloud contamination, uncertainties in vertical aerosol distribution and relatively coarse spatial resolution. These limitations can restrict their application for localized air quality assessment and sustainability-focused urban analysis, where fine-scale spatial detail is essential.

Recent advances in high-resolution Earth observation have opened new opportunities for data-driven air quality monitoring. The Sentinel-2 satellite constellation provides optical imagery at spatial resolutions as fine as 10 meters, capturing detailed information about land cover, urban structure and surface reflectance. Such surface characteristics are closely linked to emission sources, dispersion patterns and human activity and therefore serve as indirect indicators of $PM_{2.5}$ variability. Importantly, these features are also strongly connected to urban form, land-use decisions and sustainability-oriented planning policies, enabling the exploration of visual proxies for air pollution that complement traditional atmospheric measurements.

In this thesis, I investigate a spatiotemporal learning framework for estimating $\text{PM}_{2.5}$ concentrations using high-resolution Sentinel-2 satellite imagery combined with geographic location and temporal information. To mitigate noise, missing values and short-term fluctuations in ground-based air quality measurements, $\text{PM}_{2.5}$ observations are temporally aggregated, allowing the model to focus on more stable regional and seasonal trends that are more relevant for sustainability assessment and long-term environmental planning. A neural network is trained to integrate satellite-derived image features with spatial and temporal context, capturing both spatial structure and temporal patterns in air pollution levels.

The primary contributions of this work are threefold:

- (1) The development of a scalable framework for $\text{PM}_{2.5}$ estimation using high-resolution satellite imagery that supports sustainable air quality monitoring.
- (2) The integration of spatial and temporal context to improve predictive performance beyond image-only baselines.
- (3) An empirical evaluation of the proposed approach using real-world satellite imagery paired with ground-based $\text{PM}_{2.5}$ observations.

Together, these contributions provide a foundation for generating dynamic $\text{PM}_{2.5}$ maps that can inform environmental sustainability assessment, sustainable urban planning and long-term public health decision-making.

2. Literature Review

2.1 PM_{2.5}, Public Health, and Sustainability

PM_{2.5} (particles $\leq 2.5 \mu\text{m}$) is one of the most critical urban air pollutants, driven by combustion sources such as vehicle traffic, biomass and domestic fuel burning, and industrial activities [1]. Global satellite-based and modeling assessments show persistent spatial and seasonal heterogeneity, with especially high burdens in densely populated and industrializing areas [2].

Because of their small size, PM_{2.5} particles penetrate deep into the lungs and can enter the bloodstream, triggering oxidative stress, systemic inflammation and a range of cardiopulmonary pathologies [1]. Epidemiological evidence links both short- and long-term exposure to increased risks of respiratory infections, asthma exacerbations, chronic obstructive pulmonary disease, ischemic heart disease, heart failure, stroke, lung cancer and adverse reproductive outcomes [2]. Time-series studies consistently report positive associations between daily PM_{2.5} fluctuations and hospital admissions or mortality for cardiovascular and respiratory diseases, with stronger short-term effects often observed for respiratory outcomes, and higher long-term risks for cardiovascular disease [2]. Globally, ambient PM_{2.5} is estimated to contribute to millions of premature deaths annually and substantial reductions in life expectancy, particularly in heavily polluted regions of Asia and Africa.

Conventional ground-based regulatory networks provide accurate PM_{2.5} measurements but remain spatially sparse, particularly in low- and middle-income regions. This makes it harder to accurately assess exposure levels and identify differences in health risks across populations.

Thus, key research gaps include:

- (i) insufficient monitoring in underserved regions
- (ii) limited personal and micro-environment exposure data
- (iii) the need to integrate regulatory monitors, low-cost sensors, satellite remote sensing, and atmospheric models into harmonized systems.

2.2 Limitations of Ground-Based Air Quality Monitoring

Ground-based monitoring networks, including both regulatory-grade stations and low-cost sensors, play a critical role in measuring PM_{2.5} concentrations and supporting air quality regulations. They provide reliable, real-time data that forms the foundation of exposure assessment and public health analysis. However, despite their importance, these systems face

several structural and operational limitations. These challenges highlight the need for complementary approaches such as satellite remote sensing and machine learning to improve spatial coverage, scalability and overall air quality estimation. The following subsections discuss these limitations in detail.

2.2.1 Sparse and Uneven Spatial Coverage

Regulatory-grade air monitors are expensive to install, operate and maintain. As a result, monitoring networks are usually limited in number and concentrated in major urban centers, leaving large rural and peri-urban areas unmonitored [3].

Even within cities, monitor placement can be biased toward higher-income neighborhoods which reinforces environmental injustice and misses hotspots in disadvantaged areas [2].

2.2.2 Limited Resolution and Representativeness

For monitoring networks that are sparse, researchers often rely on spatial interpolation or coarse reanalysis products to estimate PM_{2.5} levels in unmonitored areas. However, these methods can underestimate concentrations and fail to capture small-scale variations, especially near traffic corridors, industrial zones, or smaller settlements where pollution levels may be much higher.

In many regions, people live tens to hundreds of kilometers away from the nearest monitoring station. This large distance can lead to exposure misclassification, making it harder to accurately link pollution levels with health outcomes and weakening the effectiveness of targeted policy interventions [4].

2.2.3 Cost, Operational Complexity, and Data Quality

Regulatory instruments such as Beta Attenuation Monitors (BAM) and Federal Reference Methods (FRM) are expensive, require continuous power supply, and need regular calibration and expert maintenance. These requirements make large-scale expansion of monitoring networks difficult, particularly in resource-constrained settings.

Low-cost sensors offer a more affordable way to increase spatial coverage. However, they often experience systematic biases, sensitivity to humidity and temperature, sensor drift over time and particle composition-specific errors. As a result, they require strong calibration procedures and continuous quality assurance and quality control (QA/QC), which are often location specific.

Calibrating large sensor networks consistently remains technically and logistically challenging, especially in cities with diverse pollution sources and varying environmental conditions [4].

2.2.4 Implications for Sustainable Monitoring

These limitations hinder high-resolution, equitable PM_{2.5} exposure mapping and long-term trend analysis needed for SDG-aligned, sustainable air quality management. Integrated frameworks that fuse sparse ground data with satellite products, auxiliary geospatial predictors, and machine-learning models are increasingly proposed to overcome these gaps and produce spatially continuous PM_{2.5} fields [2].

2.3 Satellite-Based PM_{2.5} Estimation Approaches

Satellite approaches exploit wide-area coverage to infer near-surface PM_{2.5} from column aerosol properties and surface/meteorological context.

2.3.1 AOD-Driven Physical and Statistical Models

Early and still widely used approaches estimate surface PM_{2.5} from satellite-derived aerosol optical depth (AOD) by leveraging chemical transport models (CTMs) to relate column-integrated aerosol burden to near-surface particulate mass. A commonly adopted framework integrates multi-sensor AOD products (e.g., MODIS, MISR, and MAIAC) with GEOS-Chem simulations and applies statistical bias correction using ground monitoring data. These global fusion products typically achieve cross-validated R^2 values of approximately 0.8–0.9 at spatial resolutions around 10 km [6].

At regional scales, spatiotemporal regression models further combine AOD with meteorological variables, trace gases such as NO₂, and land-use characteristics to account for spatially varying PM_{2.5} - AOD relationships. While these approaches can achieve comparable performance ($R^2 \approx 0.7$ – 0.9), they rely on sufficiently dense ground monitoring networks for calibration. Physical formulations that explicitly link aerosol extinction, fine-mode fraction, planetary boundary layer height, and relative humidity to dry PM_{2.5} provide a more mechanistic interpretation but remain sensitive to input uncertainties and often exhibit substantial hourly-scale errors [6].

2.3.2 Machine Learning and Hybrid Fusion Models

Machine learning (ML) approaches are now widely adopted to model the nonlinear and spatiotemporally varying relationships between AOD, environmental factors, and surface PM_{2.5}. Algorithms such as random forest, gradient boosting, and neural networks generally outperform traditional regression methods at daily to monthly scales, particularly in heterogeneous urban environments and complex terrain. Ensemble and two-stage frameworks such as combining chemical transport model outputs with ML-based residual correction further improve predictive accuracy and help address missing AOD observations, often achieving R^2 values above 0.8 across daily to annual scales [5].

Recent literature emphasizes the importance of carefully designing covariates (including meteorology, land use, elevation, vegetation, and gaseous pollutants), implementing robust gap-filling strategies for incomplete AOD coverage, and applying rigorous spatial and temporal cross-validation to avoid overly optimistic performance estimates. These findings also highlight that the marginal contribution of AOD can be context-dependent, particularly when rich ancillary predictors are available [5].

2.3.3 Spatiotemporal Resolution and Application Space

State of the art satellite machine learning frameworks now routinely generate daily PM_{2.5} estimates at spatial resolutions of 1–10 km across national, continental, and global domains over multi-year periods. By integrating multi-sensor satellite observations with chemical transport models and advanced statistical or ML techniques, these systems provide spatially continuous exposure surfaces suitable for long-term analysis [6].

Such products have become foundational for large-scale epidemiological studies, air quality policy evaluation, and Sustainable Development Goal (SDG) exposure assessments, particularly in regions where ground-based monitoring networks remain sparse or unevenly distributed [6].

Model family	Core idea / inputs	Typical resolution & performance
CTM + AOD (geophysical)	Convert AOD to PM _{2.5} via CTM; global bias correction	~10 km; $R^2 \approx 0.8$ –0.9 annual
Stat. spatiotemporal	Regress PM _{2.5} on AOD + met + land use + NO ₂ , etc.	1–10 km; $R^2 \approx 0.7$ –0.9

Pure ML	RF/GBM/NN on AOD + Multi covariates	1–10 km; daily $R^2 \approx 0.6\text{--}0.85+$
Hybrid ensemble	/ CTM/physical/temporal model + ML residual or stacking	1–3 km; often best overall

Table 1. Main categories of satellite-based PM_{2.5} estimation models.

2.4 Role of High-Resolution Optical Imagery and Urban Sustainability

High-resolution (≤ 10 m, often sub-meter) optical imagery enables detailed mapping of urban morphology, green/blue infrastructure, and land use dynamics, supporting climate resilience, ecosystem services, and SDG monitoring.

2.4.1 Capturing Urban Form, Roughness, and Ventilation

Very high-resolution satellite imagery, processed within GIS platforms or cloud environments such as Google Earth Engine, enables the extraction of detailed urban morphological features including building footprints, height and surface roughness proxies, and fine-scale roughness units (e.g., 200×200 m grids). These derived metrics help characterize urban airflow regimes and identify potential pollution accumulation zones, supporting healthier and low-carbon urban planning strategies [5].

Such high-resolution image integration provides substantially improved representation of urban form compared to coarser products, making it particularly valuable for fine-scale environmental assessment and sustainable urban design applications [6].

2.4.2 Mapping Green Infrastructure and Ecosystem Services

Object-based image analysis and deep learning–based segmentation applied to high-resolution satellite imagery (e.g., WorldView, NAIP, GeoEye) enable precise delineation of urban vegetation components such as tree canopies, lawns, shrubs, and surface water bodies [5]. These methods significantly improve the accuracy of metropolitan-scale greenspace inventories and support the development of functional green typologies relevant to urban cooling, air quality mitigation, and recreational planning [6].

Recent multi-sensor and AI-driven frameworks further integrate high-, medium-, and coarse-resolution imagery to quantify vegetation structure, greenness indices, and evapotranspiration dynamics for water-sensitive urban design [6]. Advanced tree-crown mapping approaches based

on artificial intelligence also strengthen biodiversity monitoring and microclimate regulation strategies within smart-city planning contexts [6].

2.4.3 Deep Learning, Big Data and SDG Assessment

Unsupervised and supervised deep learning models applied to very high-resolution imagery (0.3–0.6 m) enable the extraction of latent “phenotypes” of built and natural environments associated with urban density, vegetation structure, and infrastructure patterns [6]. These representations provide a scalable and cost-effective framework for monitoring sustainable urban development in regions where conventional socioeconomic or environmental data remain sparse [6].

Recent large-scale datasets integrate multi-year high-resolution imagery with ancillary sources such as census statistics and nighttime lights to derive spatial indicators aligned with multiple Sustainable Development Goals (SDGs), including poverty, health, inequality, and living environment conditions [5]. Complementary resolution–performance analyses further demonstrate that sub-meter imagery substantially improves the detection of SDG-relevant urban objects, including vehicles and rooftop solar panels, strengthening fine-scale monitoring capabilities [5].

Application domain	Example outputs for planning
Ventilation & climate	Surface roughness, airflow regimes
Land take & imperviousness	Imperviousness degree, urban expansion
Green infrastructure	UGS extent, type, tree crowns, ET
SDG & socioenvironmental metrics	SDG indicators, urban phenotypes, vehicles/solar

Table 2. Major sustainability uses of high-resolution imagery.

2.4.4 Conclusion

High-resolution optical imagery, increasingly integrated with machine learning techniques, provides spatially explicit and fine-grained evidence of urban form, green infrastructure distribution, and ecosystem service patterns. Such data-driven approaches are

central to the design, monitoring, and governance of more sustainable and climate-resilient cities, particularly in contexts where conventional planning data are limited or unevenly available.

2.5 Spatiotemporal Modeling for PM_{2.5} Prediction

Spatiotemporal models are designed to characterize both the spatial distribution and temporal dynamics of PM_{2.5} concentrations by integrating emissions, meteorological drivers, and atmospheric transport processes with observational data. These frameworks explicitly account for variability across locations and time, enabling more accurate representation of pollutant behavior under changing environmental conditions.

2.5.1 Statistical and Machine-Learning Spatiotemporal Models

Large-scale satellite covariate modeling frameworks commonly employ machine learning algorithms such as random forests, gradient boosting, and generalized additive models (GAMs) to predict daily PM_{2.5} concentrations on regular spatial grids [7]. These approaches are frequently implemented within multi-stage architectures that separately model long-term temporal trends and residual spatiotemporal variability to improve stability and predictive [7].

Examples are as follows:

- China-wide random forest and ensemble systems integrating AOD, chemical transport model outputs, meteorological variables, and land-use predictors have achieved daily cross-validated R^2 values in the range of approximately 0.79 - 0.83 at ~10 km resolution, with improved performance at monthly and annual aggregation scales [8].
- In Great Britain, a 1 km four-stage random forest framework incorporating monitor fusion, AOD gap-filling, core prediction, and spatial gridding reported daily cross-validated $R^2 \approx 0.77$, with strong temporal and moderate spatial predictive skill [11].
- In Indo-Gangetic Plain, a 1 km random forest model combining AOD, water vapor, meteorology, and land-use variables explained approximately 87% of variance, with performance improving as predictions were aggregated from daily to annual scales [12].

Across these systems, key design principles include explicit modeling of temporal components (e.g., GAM-based seasonal or long-term trends), region-specific sub-models to address spatial heterogeneity, and rigorous spatial and temporal cross-validation strategies to reduce overfitting in areas with clustered monitoring networks [9].

2.5.2 Deep Learning and Graph-Based Spatiotemporal Forecasting

Deep learning frameworks explicitly encode spatial neighborhood structure and temporal dependencies to model the dynamic behavior of PM_{2.5} concentrations. Graph Convolutional Long Short-Term Memory (GC-LSTM) architectures represent monitoring stations as graph nodes, enabling the model to learn spatial diffusion processes alongside temporal memory effects, and have demonstrated improved performance over classical baselines for multi-hour forecasting horizons [9].

Similarly, hybrid graph convolution and ConvLSTM architectures integrate meteorological networks and satellite-derived inputs to capture joint spatial–temporal dependencies, supporting extended forecasting windows while improving predictive skill relative to earlier spatiotemporal deep learning approaches [10]. Decomposition-based hybrid frameworks, such as CEEMDAN combined with LSTM, ARIMA, or SVM components, further enhance performance by separating PM_{2.5} time series into intrinsic modes and applying tailored predictors to each component [9].

Overall, these deep spatiotemporal models demonstrate strength in short-term forecasting (hours to days), especially in regions supported by dense monitoring networks, although they typically require substantial data availability and computational resources for training and deployment.

2.5.3 Hybrid Physical Data-Driven Frameworks

Hybrid modeling frameworks integrate process-based chemical transport models (CTMs) or Bayesian spatiotemporal fields with machine learning to combine physical interpretability with data-driven correction. For example, a CMAQ-LSTM-RF system uses LSTM networks to learn temporal structure from CTM outputs, while a random forest integrates CTM predictions with meteorological and topographic variables to generate 1 km, hourly PM_{2.5} forecasts, substantially reducing RMSE relative to CTM simulations alone [11].

Similarly, Bayesian spatial partial differential equation (SPDE) fields fused with convolutional and deep neural networks enable the modeling of smooth spatial dependence alongside nonlinear environmental responses, improving urban PM_{2.5} prediction while supporting interpretability through SHAP-based feature attribution [10].

Overall, these hybrid approaches leverage physical process knowledge to maintain consistency with atmospheric dynamics, while machine learning components correct systematic biases and introduce fine-scale variability that traditional CTMs may not fully resolve.

Class	Core idea / strengths	Typical use
Stat/ML satellite–covariate	RF/GBM/GAM on AOD, CTM, metand land use	Long term, exposure fields
Deep spatiotemporal (RNN, GCN, ConvLSTM)	Learn spatial graphs + temporal dynamics	Short-term forecasts
Hybrid CTM/Bayesian + ML	Physical fields + ML residual/bias correction	High-res now/forecast

Table 3. Main Classes of spatiotemporal PM_{2.5} prediction models.

2.6 High Resolution Satellite Imagery for Direct PM_{2.5} Estimation

A small but growing body of work has begun estimating surface PM_{2.5} directly from high-resolution satellite imagery, rather than relying on coarser aerosol optical depth (AOD) products. These approaches typically combine RGB or multispectral imagery with limited meteorological inputs and apply deep learning or hybrid machine learning frameworks to resolve pollution variability at neighborhood scales [12].

Using daily 3 m PlanetScope RGB imagery over Beijing [12], extracted 200 × 200 m image patches centered on monitoring stations and processed them through a convolutional neural network (CNN) to learn texture and color features associated with aerosol loading and urban morphology [12]. These learned representations, combined with daily meteorological variables, were input to a random forest model, achieving mean absolute errors of approximately 10 µg/m³ and correlations near 0.9 at 200 m resolution. Notably, the learned CNN features demonstrated transferability across cities, requiring retraining only of the final regression layer.

A subsequent residual-learning framework first generated a baseline PM_{2.5} surface using meteorological predictors and random forest, then trained a CNN on PlanetScope imagery to model spatial residuals at 300 m resolution, enhancing intra-urban hotspot detection and revealing neighborhood-scale contrasts on the order of several µg/m³ [13]. Similar fine-scale performance has been demonstrated in Lucknow, where a hybrid RT-RF-CNN model integrating calibrated low-cost sensors, 3 m imagery, and basic meteorological inputs produced daily 500 m PM_{2.5} maps with spatial correlations near 0.9 and clear seasonal hotspot differentiation [14].

Other frameworks incorporate high-resolution imagery as one component within broader multi-source systems. The DeepAir architecture integrates pretrained CNN-derived image features with geographic and meteorological predictors within a gradient-boosted modeling framework to generate daily 1 km PM_{2.5} estimates, outperforming standalone deep or traditional

machine learning models [15]. At continental scales, deep learning models applied directly to Sentinel-5P radiance data have demonstrated that rich spectral information alone can capture meaningful PM signals, even without explicit meteorological inputs, when paired with appropriate encoder–decoder architectures [16].

Study Region /	Imagery & Inputs	Target Resolution	Model Framework
Beijing / Shanghai	PlanetScope 3 m RGB imagery + daily meteorology	200 m	CNN extracts image features; Random Forest predicts PM _{2.5}
Beijing & Delhi	PlanetScope 3 m RGB imagery + three meteorological variables	300 m	Random Forest baseline with CNN-based residual learning for hotspot detection
Lucknow (India)	PlanetScope 3 m RGB imagery + calibrated low-cost sensors + basic meteorology	500 m	RT-RF-CNN hybrid model producing daily city-scale PM _{2.5} maps
California	High-resolution satellite imagery + geographic predictors + detailed meteorology	1 km	Pretrained CNN feature extraction integrated with LightGBM (DeepAirframework)
Europe	Sentinel-5P radiance data (no auxiliary inputs)	Continental grid	UNet-style CNN directly mapping radiance to PM _{2.5}

Table 4. Key high-resolution PM_{2.5} models using raw imagery

Collectively, these studies demonstrate that high-resolution optical and radiance data, when coupled with CNN-based or hybrid learning frameworks, can generate spatially detailed PM_{2.5} fields and uncover fine-scale exposure gradients that are often smoothed in conventional AOD-based products.

2.7 Research Gaps and Motivation for This Thesis

Recent advances demonstrate that high-resolution spatiotemporal PM_{2.5} modeling is feasible using multi-stage hybrid frameworks and ensemble learning at large scales [6]. Graph-

based deep learning has further improved short-term prediction by explicitly encoding spatial connectivity and temporal memory [13]. Despite strong accuracy, these approaches often introduce nontrivial engineering and deployment complexity, which motivates a deliberately simpler framework in this thesis [13].

2.7.1 Gaps in PM_{2.5} Spatiotemporal Modeling

Complex, hard-to-deploy frameworks

National and regional-scale models frequently rely on multi-stage pipelines that fuse satellite AOD, CTM outputs, meteorology, and land-use covariates through ensembles or hierarchical modeling [11]. These pipelines can be computationally demanding and technically specialized, which can limit reproducibility and straightforward transfer across cities [12].

Recent spatiotemporal graph and hybrid deep models can achieve strong accuracy, but they often require careful graph construction or dynamic adjacency learning that is difficult to standardize across regions [6]. This leaves a practical gap for simpler spatiotemporal approaches that can be implemented without CTMs or graph engineering [12].

Underuse of minimal spatiotemporal encodings

Many modern methods encode space and time through learned graphs, attention mechanisms, or varying-coefficient surfaces [12]. In contrast, relatively few studies test how far one can go using minimal encodings such as coordinates and a seasonal/time index paired with straightforward spatial aggregation as the main spatiotemporal operator [12].

Evidence from direct-imagery studies suggests that patch-based models capture neighborhood texture effectively, but they primarily learn local structure rather than long-range transport or multi-day dynamics [12]. More broadly, uneven monitoring can induce systematic bias through sample imbalance, which remains a known risk even for deep models [16].

Sentinel-2 underexploited for urban PM_{2.5}

Most high-resolution PM_{2.5} frameworks still center on dense AOD products or multi-sensor fusion pipelines rather than direct use of Sentinel-2 surface reflectance for urban exposure mapping [12]. At the same time, Sentinel-2's 10 - 20 m multispectral imagery has shown promise for characterizing particulate-related events and urban aerosol signatures using relatively simple

band-based analyses [15]. This points to a clear opportunity to evaluate Sentinel-2 reflectance as a primary signal source in a lightweight, transferable modeling design [12].

2.7.2 Motivation for Sentinel-2 + Location + Time + Aggregation

Sentinel-2 provides globally consistent, free, decametric multispectral imagery that can represent urban surface structure and event-related reflectance variation at neighborhood scales [12]. Separately, large-scale spatiotemporal modeling work suggests that compact representations can be competitive when paired with robust learning and validation, without requiring bespoke graph construction per city [12].

In the direct imagery $PM_{2.5}$ literature, high-resolution mapping is feasible but remains geographically narrow and often relies on city-specific pipelines that are difficult to generalize [14]. Taken together, these observations motivate a deliberately simple and scalable framework that uses Sentinel-2 reflectance with encoded location (x, y), encoded time (t), and post-hoc aggregation as the primary spatiotemporal mechanism [16]. This thesis evaluates how far such a minimal design can approximate the spatial detail and predictive performance of more complex hybrid and graph-based approaches while remaining transparent and transferable [10].

2.7.3 Research Questions

RQ1 – Image vs. Spatiotemporal Models

To what extent does incorporating spatial (location) and temporal (date/season, aggregated $PM_{2.5}$) context alongside Sentinel-2 imagery improve $PM_{2.5}$ prediction performance compared with image-only models?

RQ2 – Role of Temporal Aggregation

How does temporal aggregation of ground-based $PM_{2.5}$ measurements (e.g., daily/weekly/monthly averages) affect model accuracy and stability in capturing regional and seasonal pollution patterns?

3. METHODOLOGY

3.1 Problem Formulation

This thesis formulates $\text{PM}_{2.5}$ estimation as a supervised regression problem. The objective is to predict near-surface $\text{PM}_{2.5}$ concentrations at a given geographic location and time using satellite image patches together with compact spatial and temporal descriptors. Similar supervised learning frameworks are widely used in satellite-based $\text{PM}_{2.5}$ modeling, where ground observations serve as labels and remotely sensed or geospatial variables act as predictors [16].

To emphasize stable pollution patterns relevant for sustainability assessment and long-term planning, ground-based $\text{PM}_{2.5}$ observations are aggregated to weekly means. Weekly or longer aggregations are commonly adopted to smooth measure noise, reduce missingness, and highlight persistent spatial and seasonal variability [11]. Such aggregation also aligns with evidence demonstrating clear weekly and seasonal $\text{PM}_{2.5}$ cycles and the importance of synoptic-scale patterns for exposure assessment [13].

Let $i \in \{1, \dots, N\}$ index monitoring locations and $t \in \{1, \dots, T\}$ index weekly time periods. For each location time pair (i, t) , the dataset consists of:

- A satellite image patch $\mathbf{X}_{i,t} \in \mathbb{R}^{H \times W \times C}$, centered at location i , representing surrounding surface and land-cover characteristics.
- A spatial feature vector $\mathbf{s}_i$, containing geographic information of latitude and longitude.
- A temporal feature vector $\boldsymbol{\tau}_t$, encoding time-related information which is the week of the year.
- A target value $y_{i,t}$, denoting the weekly averaged $\text{PM}_{2.5}$ concentration measured at location i during week t .

The learning task is to estimate a function $f(\mathbf{X}_{i,t}, \mathbf{s}_i, \boldsymbol{\tau}_t) \rightarrow y_{i,t}$, where $f(\cdot)$ is a parameterized model trained to minimize the discrepancy between predicted and observed $\text{PM}_{2.5}$ values across all samples.

Weekly aggregation reduces short-term sensor noise, missing observations, and transient pollution spikes, allowing the model to focus on stable spatial and seasonal patterns rather than high-frequency fluctuations [12]. This formulation prioritizes generalization across locations and time periods over short-term forecasting accuracy, aligning with the broader objective of scalable, sustainability-oriented air quality mapping (He & Huang, 2018).

3.2 Data Sources

This study integrates ground-based PM_{2.5} observations, satellite remote sensing imagery, and spatial-temporal metadata to construct a unified dataset for supervised PM_{2.5} estimation. All data sources are spatially aligned at monitoring locations and temporally aligned at a weekly resolution.

3.2.1 Ground-Based PM_{2.5} Measurements (OpenAQ)

Ground-truth near-surface PM_{2.5} concentrations are obtained from the OpenAQ platform (version 3), which aggregates regulatory and reference-grade air quality measurements from government and research monitoring networks worldwide. OpenAQ provides standardized access to PM_{2.5} mass concentration measurements reported in micrograms per cubic meter ($\mu\text{g}/\text{m}^3$).

Daily PM_{2.5} observations from individual monitoring stations were temporally aggregated into weekly averages. This aggregation reduces high-frequency sensor noise, mitigates the impact of missing daily measurements, and emphasizes more stable regional and seasonal pollution patterns. Stations with insufficient temporal coverage or excessive missing data were excluded to ensure the reliability and consistency of weekly aggregated PM_{2.5} labels used for model training and evaluation.

3.2.2 Satellite Remote Sensing Data (Sentinel-2)

Satellite imagery was sourced from the Sentinel-2 mission operated by the European Space Agency (ESA). Specifically, Sentinel-2 Level-2A surface reflectance products were used, which provide atmospherically corrected optical imagery suitable for land-surface analysis.

This study utilizes visible-band imagery (Bands B4, B3, and B2 corresponding to red, green, and blue wavelengths) at a spatial resolution of 10 meters. These bands capture fine-scale surface and land-cover characteristics, including urban structure, vegetation, and surface reflectance patterns that serve as indirect proxies for PM_{2.5} variability. For each monitoring station and week, square satellite image patches centered on the station coordinates were extracted to represent the surrounding local environment, influencing air pollution levels.

All Sentinel-2 imagery was accessed and processed using the Google Earth Engine platform, which enables large-scale geospatial data retrieval, filtering, and export. Only images corresponding to the temporal window of each weekly PM_{2.5} aggregation were considered to ensure temporal consistency between satellite observations and ground-based measurements.

3.2.3 Spatial and Temporal Contextual Features

In addition to satellite imagery, spatial and temporal contextual features were incorporated as auxiliary model inputs. Spatial features consist of the geographic coordinates (latitude and longitude) of each monitoring station, enabling the model to learn region-specific background pollution characteristics and broad spatial gradients in $\text{PM}_{2.5}$ concentrations.

Temporal features encode the timing of each observation, including the corresponding week of the year, allowing the model to capture seasonal and cyclical variability in $\text{PM}_{2.5}$ driven by factors such as meteorology, emission patterns, and human activity. These spatial and temporal features complement satellite-derived image representations by providing contextual information that may not be fully captured by visual surface characteristics alone.

3.3 Data Preprocessing

A series of preprocessing steps were applied to ensure data quality, temporal consistency, and spatial alignment across ground-based $\text{PM}_{2.5}$ observations, satellite imagery, and auxiliary contextual features. All preprocessing operations were conducted at a weekly temporal resolution and at the spatial scale defined by station-centered satellite image patches.

3.3.1 Temporal Aggregation and Label Construction

Daily $\text{PM}_{2.5}$ mass concentration measurements obtained from OpenAQ were aggregated into weekly mean concentrations for each monitoring station. Weekly aggregation was employed to (i) reduce the impact of high-frequency measurement noise, (ii) mitigate the effects of intermittent missing daily observations, and (iii) suppress short-term episodic variability that is not representative of longer-term exposure patterns. The resulting weekly $\text{PM}_{2.5}$ values serve as the regression targets for model training and evaluation.

All auxiliary data sources were temporally aligned to the same weekly intervals. For each station-week pair (i, t) , only satellite observations acquired within the corresponding aggregation window were considered, ensuring temporal consistency between predictor variables and target labels.

3.3.2 Satellite Image Quality Control and Cloud Masking

Satellite imagery was obtained from Sentinel-2A surface reflectance products and processed using the Google Earth Engine. Level-2A products include a scene classification layer

that identifies clouds, cloud shadows, cirrus, and other invalid pixels. These quality flags were used to generate binary masks that exclude contaminated pixels from subsequent analysis.

For each monitoring station and week, square image patches were extracted from the visible spectral bands (B4, B3, and B2). To ensure sufficient informative content within each patch, samples exhibiting excessive cloud contamination or a low proportion of valid pixels were removed from the dataset. This filtering step reduces noise introduced by atmospheric artifacts and prevents degraded image samples from adversely affecting model learning.

3.3.3 Spatial Alignment and Patch Extraction

Monitoring station locations were represented using geographic coordinates expressed in a consistent spatial reference system. Sentinel-2 imagery was resampled to a uniform spatial resolution of 10 meters to maintain consistency across all samples. Fixed-size square image patches centered on each station coordinate were extracted to represent the local surface environment surrounding the monitoring location.

This station-centered extraction strategy is designed to capture land-cover composition, urban structure, and surface reflectance characteristics most relevant to near-surface PM_{2.5} variability, while minimizing the inclusion of unrelated distant features. Spatial alignment between station coordinates and satellite imagery was verified to ensure accurate correspondence between ground-based measurements and extracted image content.

3.3.4 Sample Filtering and Normalization

Monitoring stations with insufficient temporal coverage or excessive missing weekly PM_{2.5} observations were excluded to ensure the reliability of supervisory signals during model training. This filtering step improves dataset consistency and reduces bias introduced by sparsely sampled locations.

Prior to model input, all feature groups were normalized to facilitate stable optimization during neural network training. Satellite image reflectance values were scaled to a consistent numeric range, while spatial and temporal features were standardized to comparable magnitudes. These normalization procedures reduce scale imbalance between heterogeneous inputs and improve convergence behavior during training.

3.4 Feature Construction

Feature construction was performed to represent satellite imagery, spatial context, and temporal context in a form suitable for supervised learning. Each sample corresponding to a monitoring location i and week t is represented by a structured set of input features that jointly capture local surface characteristics and broader spatiotemporal patterns influencing $\text{PM}_{2.5}$ variability.

3.4.1 Satellite Image Features

For each station - week pair (i, t) , a satellite image patch $\mathbf{X}_{i,t} \in \mathbb{R}^{H \times W \times C}$ was constructed, where H and W denote the spatial dimensions of the patch and C denotes the number of spectral channels. In this study, $C = 3$, corresponding to the red, green, and blue visible bands. These channels capture surface reflectance characteristics associated with land cover, vegetation, and built environments, which serve as indirect indicators of emission sources and dispersion behavior.

The image tensor $\mathbf{X}_{i,t}$ constitutes the primary high-dimensional input to the model and is processed using convolutional layers to extract compact feature representations. This representation learning enables the model to identify spatial patterns in surface structure and land-use composition that are correlated with $\text{PM}_{2.5}$ concentrations.

3.4.2 Spatial Features

Spatial context was represented using a low-dimensional feature vector $\mathbf{s}_i = [\text{lat}_i, \text{lon}_i]$, where lat_i and lon_i denote the geographic latitude and longitude of monitoring station i . These features allow the model to learn broad spatial gradients in $\text{PM}_{2.5}$ concentrations, such as regional background pollution levels and large-scale geographic variability.

Spatial features were treated as auxiliary inputs and concatenated with image-derived features during model training. This explicit encoding of location information enables the model to differentiate visually similar environments located in different geographic regions, thereby improving generalization across space.

3.4.3 Temporal Features

Temporal context was encoded using a feature vector $\boldsymbol{\tau}_t$, which represents the timing of each observation. In this study, temporal features include the week of the year corresponding to

week t , enabling the model to capture seasonal patterns in $\text{PM}_{2.5}$ concentrations driven by factors such as meteorology, emission cycles, and human activity.

Incorporating temporal features allows the model to distinguish between similar surface conditions observed at different times of the year and to learn recurring seasonal trends. This temporal encoding is particularly important for $\text{PM}_{2.5}$, which exhibits strong intra-annual variability.

3.4.4 Feature Integration

The final model input for each sample is formed by integrating the three feature groups: $(\mathbf{X}_{i,t}, \mathbf{s}_i, \boldsymbol{\tau}_t)$. Satellite image features capture fine-scale local surface characteristics, while spatial and temporal features provide broader contextual information that is not fully observable from imagery alone. By combining these complementary inputs, the model is able to represent both localized environmental conditions and large-scale spatiotemporal patterns influencing $\text{PM}_{2.5}$ variability.

This feature construction strategy supports a flexible and scalable learning framework that balances high-resolution spatial detail with interpretable contextual information, aligning with the objective of sustainable air quality monitoring.

Feature Group	Symbol	Description	Purpose
Satellite Image Patch	$\mathbf{X}_{i,t}$	Station-centered Sentinel-2 RGB image patch representing local surface and land-cover characteristics	Captures fine-scale spatial patterns related to urban structure, vegetation, and surface reflectance
Spatial Features	$\mathbf{s}_i$	Geographic coordinates of monitoring station i (latitude, longitude)	Encodes regional background pollution levels and large-scale spatial variability
Temporal Features	$\boldsymbol{\tau}_t$	Temporal indicators corresponding to week t of year	Captures seasonal and cyclical variability in $\text{PM}_{2.5}$ concentrations
Target Variable	$y_{i,t}$	Weekly averaged near-surface $\text{PM}_{2.5}$ concentration	Supervision signal for model training and evaluation

Table 5. Summary of Input Features Used for PM_{2.5} Estimation

3.5 Model Architecture and Training

This study employs a neural network architecture designed to jointly process high-resolution satellite imagery and low-dimensional spatiotemporal contextual features for PM_{2.5} estimation. The model integrates convolutional feature extraction with auxiliary inputs to capture both localized surface characteristics and broader spatial and seasonal patterns influencing air pollution.

3.5.1 Model Architecture

The model consists of three main components: (i) an image feature extractor, (ii) a contextual feature encoder and (iii) a regression head.

Satellite image patches $X_{i,t}$ are processed through a convolutional neural network (CNN) to extract compact feature representations from high-dimensional image inputs. Convolutional layers are used to learn spatial patterns related to land cover, urban structure, and surface reflectance, followed by pooling operations to reduce spatial dimensionality and improve translational robustness.

Spatial and temporal features (s_i, τ_t) are processed through a separate fully connected pathway to encode geographic and seasonal context. These contextual embeddings capture large-scale background pollution gradients and periodic temporal behavior that are not fully observable from imagery alone.

To maintain simplicity and scalability, this thesis adopts a modular architecture consisting of two components:

- A pretrained convolutional neural network (CNN) for image feature extraction.
- A standard tabular regression model for prediction.

This approach avoids designing a custom deep architecture and instead reuses validated components that have been successfully applied in satellite-based PM_{2.5} modeling [12].

Two configurations are evaluated:

- **Image-only model:** CNN features only.
- **Fused model:** CNN features + location + time features.

This allows assessment of how much predictive value is added by simple spatial and temporal encoding.

3.5.2 Baseline Model

To quantify the contribution of spatiotemporal context, an image-only baseline model is trained using the same convolutional architecture but excluding spatial and temporal inputs. This baseline enables a controlled comparison between models that rely solely on satellite imagery and those that integrate additional contextual information, isolating the effect of spatial and temporal feature inclusion.

3.5.3 CNN Feature Extraction

A pretrained CNN backbone (ResNet-18) is used to extract features from Sentinel-2 image patches. The network is initialized with ImageNet weights, and the final classification layer is removed. Global average pooling is applied to produce a fixed-length feature vector for each image patch.

Only the RGB-equivalent Sentinel-2 bands (B4, B3, B2) are used as input to align with pretrained architectures. The backbone is either frozen or lightly fine-tuned during training.

Formally, for each location–time pair (i, t) : $z_{\{i,t\}} = g(X_{\{i,t\}})$ where $X_{\{i,t\}}$ is the image patch and $z_{\{i,t\}}$ is the extracted feature embedding.

3.5.4 Feature Fusion and Regression

The CNN embedding is concatenated with simple spatial and temporal features:

- Latitude
- Longitude
- Week-of-year encoding

This produces a combined feature vector: $h_{i,t} = [z_{i,t}, s_i, \tau_t]$. A standard tabular regression model (e.g., LightGBM or Random Forest) is then trained to predict weekly $\text{PM}_{2.5}$ concentrations:

$$y_{i,t} = r(h_{i,t}),$$

where $r(\cdot)$ minimizes prediction error over the training set. Gradient-boosted trees are widely used in $\text{PM}_{2.5}$ mapping due to their robustness and strong performance on heterogeneous environmental predictors [4].

This proposed architecture is intentionally simple, scalable, and transferable across cities. By avoiding reliance on chemical transport models, graph-based spatial engineering, or highly customized deep architectures, the framework prioritizes reproducibility, computational

efficiency, and ease of implementation. A pretrained CNN backbone will be used to extract high-resolution surface features from Sentinel-2 imagery, while minimal spatial and temporal encodings will be incorporated to capture persistent geographic and seasonal structure without complex spatiotemporal design.

This work will evaluate whether such a compact supervised learning framework can approximate the spatial detail and predictive performance of more complex hybrid or graph-based models, while remaining transparent and deployable across heterogeneous urban settings. The goal is to develop a scalable approach to fine-scale $\text{PM}_{2.5}$ mapping that supports environmental assessment and sustainable urban planning.

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